

Egreen Quanta

Quantum-Inspired Fuel Prediction & Green Fleet Optimization — Technical Report

SIH Problem Statement 2. This report summarizes the prediction model, the optimization algorithms (classical GA, NSGA-II, and QIGA), their benchmarked performance, and scenario analysis, generated directly from the project's saved experiment results.

1. Fuel Consumption Prediction Model

An XGBoost regressor was trained on a 9,000-voyage synthetic dataset generated from the Admiralty formula, calibrated against real EU MRV 2024 fleet data (12,957 ships) and Kaggle fuel-efficiency data.

Metric	Value
R-squared	0.90
MAPE	23%
Training set size	9,000 voyages (300 vessels x 30 voyages)
Target transform	log1p(fuel_kg) — corrects for orders-of-magnitude scale spread

2. Optimization Algorithm Comparison

Three algorithms were implemented and benchmarked on identical fleet data: a classical Genetic Algorithm (GA), NSGA-II (multi-objective), and a Quantum-Inspired Genetic Algorithm (QIGA) built from qubit-pair encoding, measurement/collapse, and a rotation-gate update.

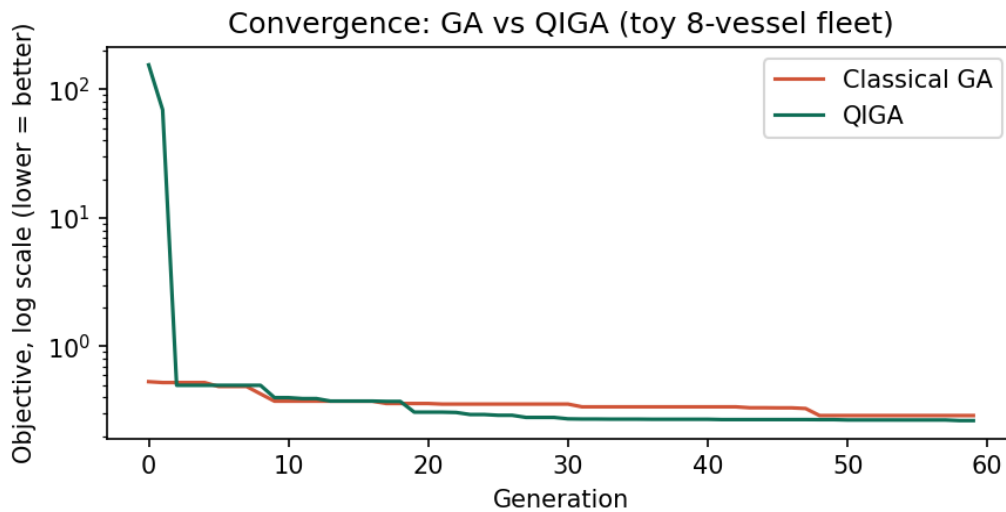


Figure 1: QIGA converges faster and to a better objective than classical GA on the toy 8-vessel fleet.

3. Statistical Validation (Multi-Seed Benchmark)

A single-seed comparison is not statistically meaningful. Both algorithms were run over 15 independent random seeds. QIGA beat GA in 12/15 seeds (80%). A paired t-test gives $p = 0.0015$, a statistically significant result at $p < 0.05$.

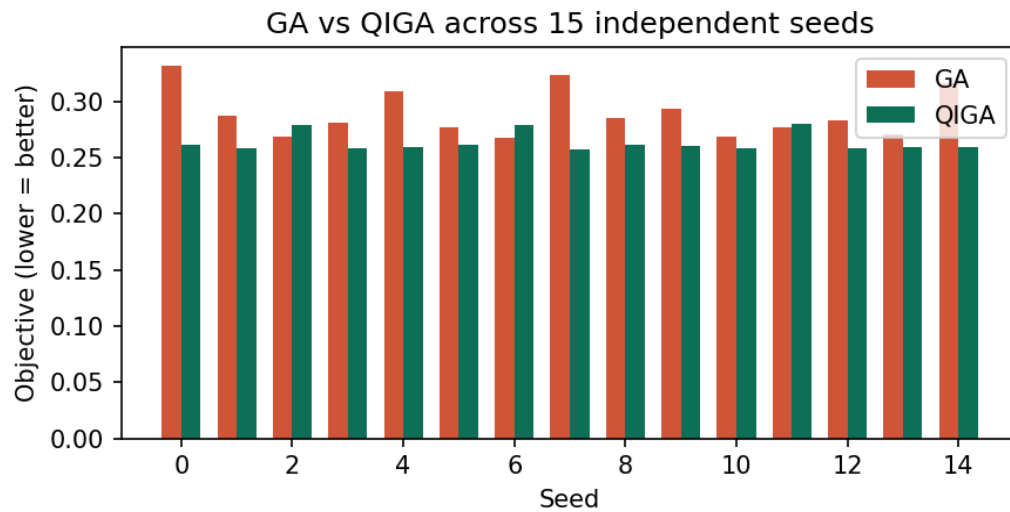


Figure 2: Per-seed objective values, GA vs QIGA.

4. Scale-Up Results

The optimization problem was scaled from the 8-vessel toy fleet up to 200, 500, 1000, and 1440 vessels (matching the full size of the reference Kaggle fleet dataset). Two structural constraint-handling issues were found and fixed at scale: a fuel-adoption-cap repair (random assignment satisfies all caps with near-zero probability at 50+ vessels) and a closed-form CII speed repair (the original flat fleet-average CII threshold breaks once Container Ships, with real fuel-per-nm around 30x a Bulk Carrier's, enter the mix).

Vessels	GA feasible?	QIGA feasible?	QIGA objective	QIGA runtime
8	Yes	Yes	0.26	0.2s
50	No	Yes	0.27	6.4s
200	No	Yes	0.31	25.6s
500	not tested	Yes	0.35	59.6s
1000	not tested	Yes	0.40	119.3s
1440	not tested	Yes	0.40	170.1s

GA was not retested past 200 vessels since it was already failing to find any feasible solution there with matched search budget — QIGA's scaling advantage is the reported finding, not an oversight.

5. Scenario Analysis

Three fuel-adoption scenarios were run through the full optimization pipeline on the toy 8-vessel fleet, varying the maximum permitted adoption rate of each fuel type (reflecting different assumptions about bunkering infrastructure availability).

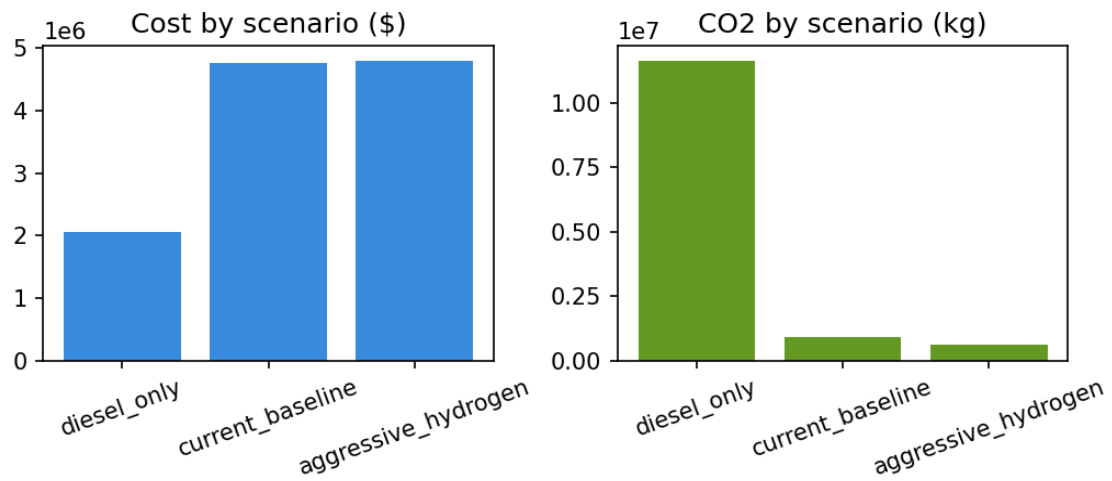


Figure 3: Cost/CO2 trade-off across diesel-only, current-baseline, and aggressive-hydrogen fuel-adoption scenarios.